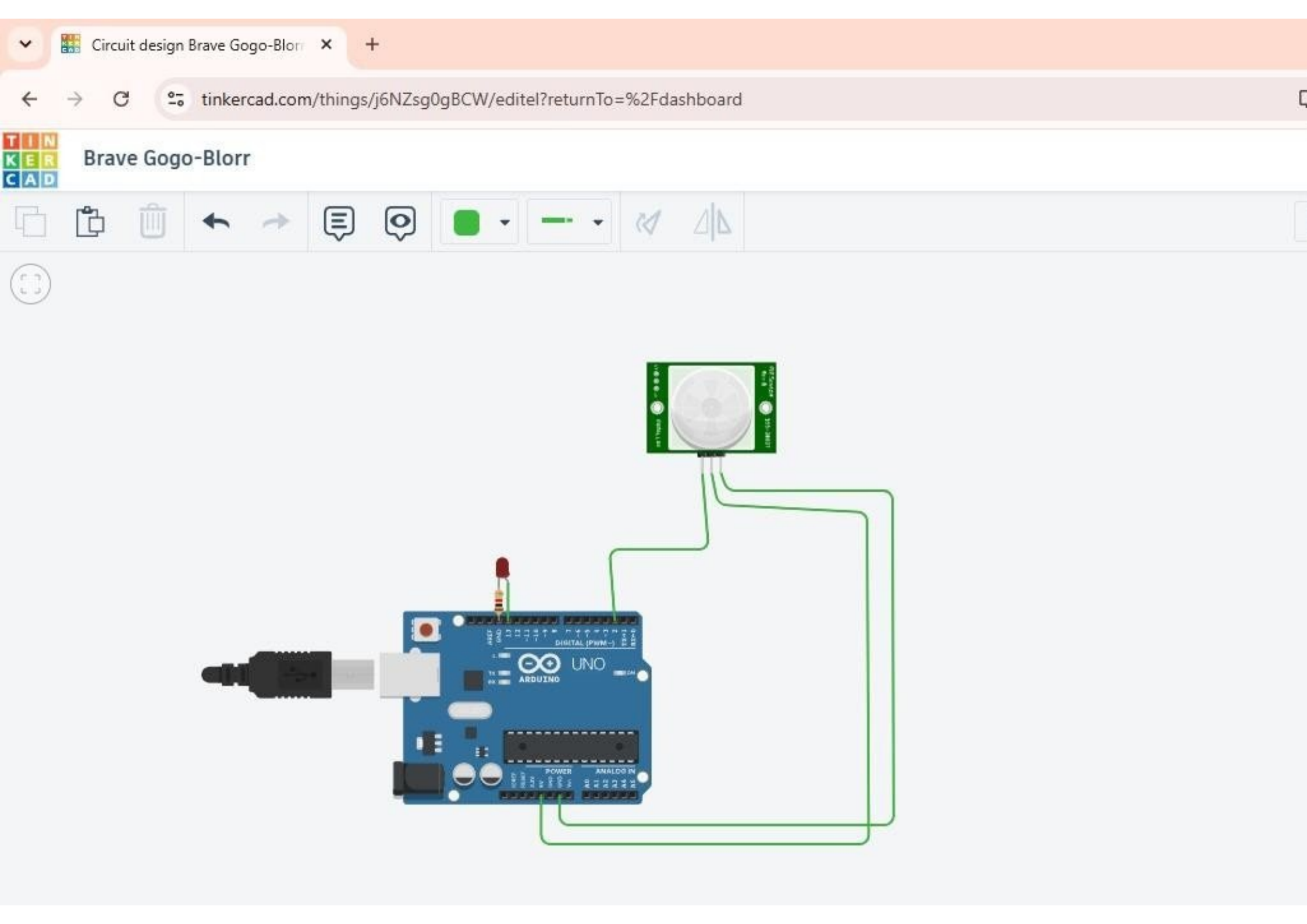
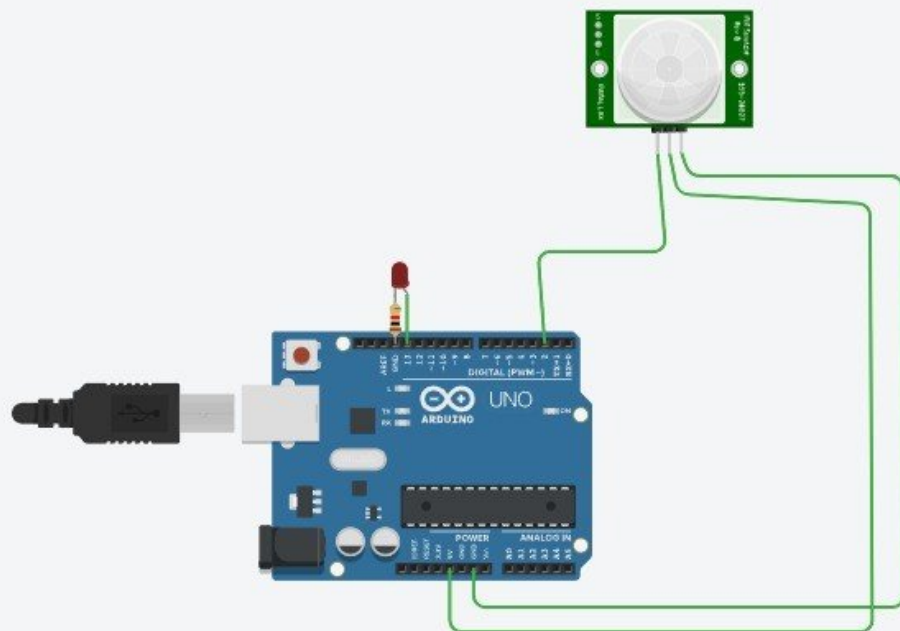
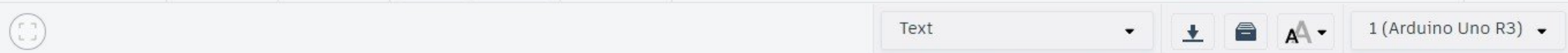
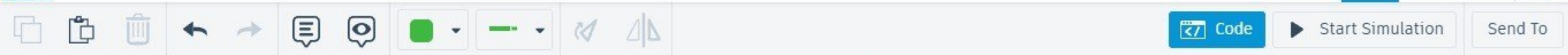


Overview of Motion Detection Alarm

The **Motion Detection Alarm** is an embedded systems project that uses an **Arduino Uno** and a **PIR (Passive Infrared) sensor** to detect human motion. The PIR sensor senses changes in infrared radiation emitted by the human body. When motion is detected, the Arduino processes the sensor signal and activates a **buzzer** (and an LED, if connected) to alert the user. When no motion is detected, the alarm remains off.

This project demonstrates the practical use of **sensor interfacing, Arduino programming, and real-time motion detection**. It is commonly used in **home security systems, offices, smart buildings, and automatic lighting systems** to improve safety and security.

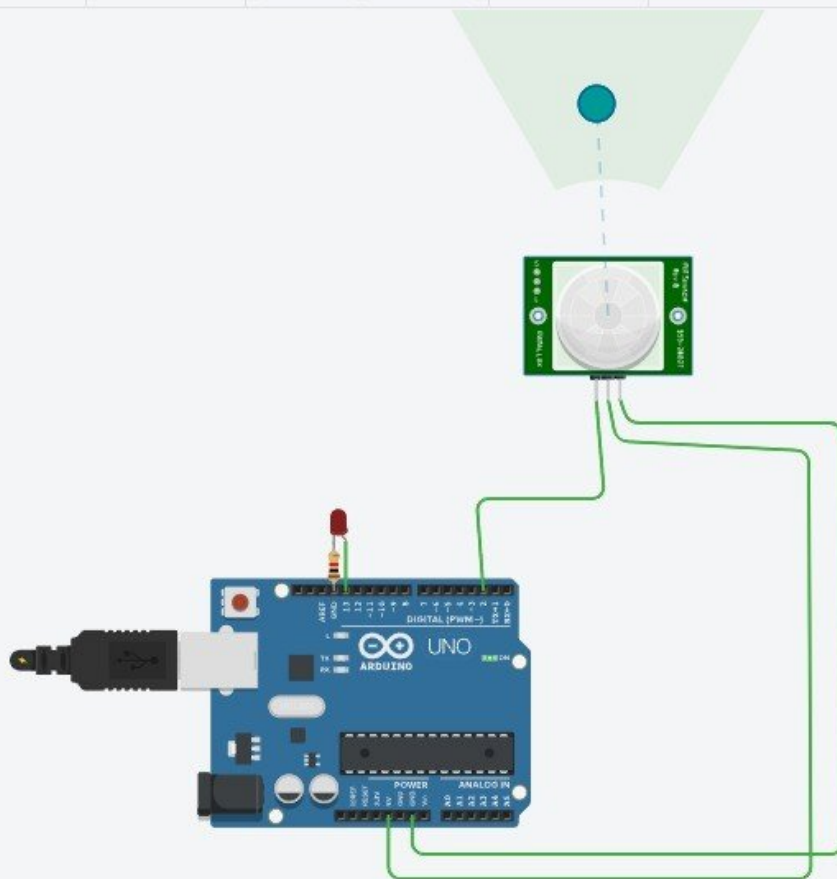




Text     1 (Arduino Uno R3) 

```
1 int buttonState=0;
2
3 void setup()
4 {
5     pinMode(2,INPUT);
6     pinMode(LED_BUILTIN,OUTPUT);
7 }
8 void loop()
9 {
10     buttonState=digitalRead(2);
11     if (buttonState == HIGH)
12     {
13         digitalWrite(LED_BUILTIN,HIGH);
14     }
15     else
16     {
17         digitalWrite(LED_BUILTIN,LOW);
18     }
19     delay(10);
20 }
```

Serial Monitor

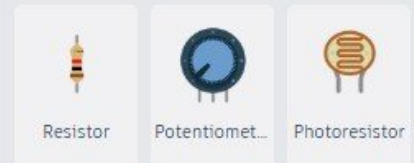


PIR Sensor

Name 1

Start / stop simulation
Spacebar

resistor



Other Components



Starters



CODE EXPLANATION:

```
int buttonState = 0;
```

declares an integer variable named buttonState.
initially ,its value is 0.

```
void setup()
```

setup() runs only once when the Arduino starts.

pin mode(2,input); sets digital pin 2 as a input to receive signals from the sensor.

set the built in LED as an output.

loop() runs continuously.

```
buttonState=digitalRead(2);
```

Reads the signal from pin 2.

```
If(buttonState == HIGH)
{
Digitalwrite(LED_BUILTIN,HIGH);
}
```

If the input is high(motion detected or button pressed),the built in LED turns ON.

If the input is LOW(no motion or button not pressed),the built in LED turns OFF.

Delay(10);

Wait for 10 milliseconds before checking the input again.